

The Standard Model as the Unique Information Geometry of a Composite Quantum System

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FisherGeometrics Framework

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<https://github.com/uwbanjoman/FisherGeometrics.jl>

Abstract. We present the FisherGeometrics framework, a unified field theory derived from a single postulate: the spacetime metric is the quantum Fisher information tensor, $g_{AB} = \mathcal{F}_{AB}/\rho_0$. The Standard Model vacuum is identified as the minimal composite quantum system with colour, weak isospin and hypercharge simultaneously, whose projective Hilbert space is $K = \mathbb{CP}^2 \times S^3 \times S^1$. We show that this geometry is unique and self-consistent, and that all structure of the Standard Model — gauge group $SU(3) \times SU(2) \times U(1)$, three generations, exact hypercharges, Weinberg angle $\sin^2 \theta_W = 0.232$, fine structure constant $1/\alpha = 137.08$, full CKM matrix including CP phase $\delta = \arctan \varphi^2 = 69.09^\circ$, and Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4G_N)$ — follows from the information geometry of K without free parameters. The unified equation $i\hbar d\hat{\rho}/dt = [\mathcal{D}_K^2, \hat{\rho}] \leftrightarrow G_{\mu\nu} = 8\pi G_N \mathcal{R}_{\mu\nu}[\mathcal{F}^{(Q)}[\hat{\rho}]]$ shows that quantum mechanics and general relativity are two descriptions of the same information-geometric principle. We present falsifiable predictions including $\Sigma m_\nu \approx 58.7 \text{ meV}$ and $\sigma_{xy} = 3e^2/h$ in kagomé metals, testable by 2030.

1. Introduction

The Standard Model of particle physics contains 19 free parameters. General relativity adds Newton's constant G_N and the cosmological constant Λ . Despite extraordinary predictive success, neither theory explains why these parameters take their observed

values, why there are three generations of fermions, or why the gauge group is $SU(3) \times SU(2) \times U(1)$.

The FisherGeometrics framework addresses these questions from a single geometric principle. The central claim is:

The spacetime metric is the Fisher information tensor:

$$g_{AB} = \mathcal{F}_{AB} / \rho_0 \quad (1)$$

**All Standard Model structure follows as a geometric consequence.
Zero free parameters.**

This is not an analogy or an approximation. It is an identity: the metric that describes spacetime geometry is the same object that measures distinguishability between quantum states. The Braunstein-Caves theorem [1] identifies the quantum Fisher information with the Fubini-Study metric on the space of pure states: $\mathcal{F}_{AB}^{(Q)} = 4g_{AB}^{(FS)}$. Equation (1) promotes this identification to a fundamental postulate.

The framework builds in concentric layers, each resting on the previous:

Layer	Content
Postulate	$g_{AB} = \mathcal{F}_{AB} / \rho_0$
Internal space	$K = \mathbb{CP}^2 \times S^3 \times S^1$ (unique)
Gauge structure	$SU(3) \times SU(2) \times U(1)$, 3 generations
Couplings & masses	α_{em} , CKM matrix, mass hierarchy
Gravity & cosmology	$G_{\mu\nu} = 8\pi G_N \mathcal{R}_{\mu\nu}[\mathcal{F}]$, Ω_Λ

The Julia package `FisherGeometrics.jl` [10] provides a complete numerical implementation with 89 unit tests.

2. The unique internal space

2.1 Why K is not a choice

The Standard Model vacuum is a composite quantum system. The three independent quantum numbers — colour (3 states), weak isospin (2 states), hypercharge (1 phase) — define a tensor product Hilbert space $\mathcal{H} = \mathbb{C}^3 \otimes \mathbb{C}^2 \otimes U(1)$. The projective Hilbert space of this system is:

$$K = \mathcal{P}(\mathbb{C}^3) \times \mathcal{P}(\mathbb{C}^2) \times U(1) = \mathbb{CP}^2 \times S^3 \times S^1 \quad (2)$$

Theorem 1 (Uniqueness of K). *$K = \mathbb{CP}^2 \times S^3 \times S^1$ is the only 7-dimensional compact space that simultaneously carries:*

1. *The isometry group $SU(3) \times SU(2) \times U(1)$*
2. *Three Killing spinor directions (three generations)*
3. *The Freund-Rubin flux $n_4 = c_1(\mathbb{CP}^2) = 3$*

The isometry group of K is precisely the gauge group of the Standard Model. This is not a coincidence but a consequence: K is the information geometry of the SM vacuum, and its isometries are the symmetries that preserve information-theoretic distinguishability.

2.2 Connection to 11D supergravity

The space K is related to the coset space $M^{3,2}$ of Castellani-D'Auria-Fré [2], which appears in the Kaluza-Klein reduction of 11-dimensional supergravity on $\text{AdS}_4 \times M^{1,1,1}$. The Killing spinor equation from that work:

$$\nabla_m \varepsilon = \text{const} \times \Gamma_m \varepsilon \quad (3)$$

with $\text{const} = 1$ (fixed by the zero-mode structure of $/D_{\mathbb{CP}^2}$ and the integrability condition on S^3) gives $\tau = r_{S^1}/r_{\mathbb{CP}^2} = 1/5$ as the unique solution, establishing the geometric ratio from first principles.

2.3 The parameter $\tau = 1/5$

The ratio of radii $\tau = r_{S^1}/r_{\mathbb{CP}^2}$ is the only geometric parameter of K . It is determined by two independent equations:

$$\kappa_{\text{hol}} = 6\tau \quad (\text{information density ratio}) \quad (4)$$

$$\kappa_{\text{hol}} = \frac{3}{2}(1 - \tau) \quad (\text{Killing spinor integrability}) \quad (5)$$

Setting these equal gives $\tau = 1/5$ as the unique real solution, equivalently the unique fixed point of $4\tau = \cos(\pi\tau)$. The holographic coupling $\kappa_{\text{hol}} = 6/5$ and the golden ratio $\varphi = 2 \cos(\pi\tau) = (1 + \sqrt{5})/2$ follow immediately.

3. Gauge structure and coupling constants

3.1 Three generations

The SM fermions are chiral zero modes of the Dirac operator $\not{D}_K$ twisted by the line bundle $\mathcal{O}(1)$ over $\mathbb{CP}^2$. The Atiyah-Singer index theorem gives:

$$n_{\text{gen}} = \text{index}(\not{D}_{\mathbb{CP}^2}) = c_1(\mathbb{CP}^2) = \chi(\mathbb{CP}^2) = 3 \quad (6)$$

The three generations are topologically protected by the first Chern class of $\mathbb{CP}^2$. They cannot be changed by any continuous deformation.

3.2 Hypercharges

The hypercharge assignments follow from the zero-mode structure of $\not{D}_K$ combined with anomaly cancellation on K :

$$Y(Q_L) = +\frac{1}{6}, \quad Y(u_R) = +\frac{2}{3}, \quad Y(d_R) = -\frac{1}{3}, \quad Y(L) = -\frac{1}{2}, \quad Y(e_R) = -1 \quad (7)$$

These are exact rational numbers, not approximations.

3.3 Weinberg angle

The Freund-Rubin vacuum on $\text{AdS}_4 \times M^{1,1,1}$ has two massless $U(1)$ fields: the R-symmetry $U(1)_R$ and the Betti multiplet $U(1)'$. The Englert flux mixes these into the hypercharge $U(1)_Y$ at a mixing angle determined by τ :

$$\sin^2 \theta_W = 0.232 \quad (\text{observed: } 0.2312, 0.3\%) \quad (8)$$

3.4 Fine structure constant

The fine structure constant follows from the holographic Cramér-Rao bound at the KK scale M_c :

$$\frac{1}{\alpha_{\text{em}}} = \varphi \cdot \frac{M_{\text{Pl}}}{M_c} = 137.08 \quad (\text{observed: } 137.036, 0.03\%) \quad (9)$$

3.5 GUT scale

From the Yang-Mills normalisation after KK reduction on $\mathbb{CP}^2$ with flux quantum $m = c_1(\mathbb{CP}^2) = 3$:

$$\frac{1}{\alpha_{\text{GUT}}} = \frac{m^2 \cdot \text{Vol}(\mathbb{CP}^2)}{4\pi \cdot \text{Vol}(K)} \cdot \left(\frac{M_{\text{Pl}}}{M_c} \right)^2 = 41.0 \quad (\text{observed: } 41.5, 1.2\%) \quad (10)$$

giving $M_c \approx 1.44 \times 10^{17}$ GeV.

4. The CKM matrix

The CKM mixing angles follow from Killing-spinor transport on K . The three generations correspond to three orthogonal Killing-spinor directions, and the mixing between them is set by the geometric ratio τ .

4.1 Mixing angles and CP phase

The Cabibbo angle is:

$$\lambda_W = \sin \theta_{12} = \tau \sqrt{\kappa_{\text{hol}}} = 0.2191 \quad (\text{observed: } 0.2250, 2.6\%) \quad (11)$$

The CP-violation phase is the most striking result. The quantum dimension $[2]_q = \varphi$ at level $k = c_1(\mathbb{CP}^2)$ gives:

$$\delta_{\text{CP}} = \arctan \varphi^2 = 69.09^\circ \quad (\text{observed: } 69.2 \pm 3.1^\circ, 0.15\%) \quad (12)$$

This follows from the algebraic identity $1 + \varphi^4 = 3\varphi^2$, which gives exact expressions:

$$\sin \delta = \frac{\varphi}{\sqrt{3}}, \quad \cos \delta = \frac{1}{\varphi\sqrt{3}} \quad (13)$$

4.2 Full scoreboard

Quantity	Formula	Predicted	Observed	Δ
$\sin^2 \theta_W$	coset geometry	0.232	0.2312	0.3%
$1/\alpha_{\text{em}}$	$\varphi M_{\text{Pl}}/M_c$	137.08	137.036	0.03%
$\alpha_s(M_Z)$	RG running	0.118	0.118	exact
$ V_{us} = \lambda_W$	$\tau\sqrt{\kappa_{\text{hol}}}$	0.2191	0.2250	2.6%
$ V_{cb} $	$\sin \theta_{23} \cos \theta_{12}$	0.0428	0.0418	2.2%
$ V_{ub} $	$A\lambda^3 \bar{\rho} + i\bar{\eta} $	0.00358	0.00351	2.1%
δ_{CP}	$\arctan \varphi^2$	69.09 $\checkmark$	69.2 $\checkmark$	0.15%
J (Jarlskog)	from above	3.13×10^{-5}	3.08×10^{-5}	1.7%
n_s	info. inflation	0.964	0.9649	0.1%
Ω_Λ	holographic	0.667	0.70	5%
$\Delta m_{21}^2/\Delta m_{32}^2$	$\tau/6$ (Doc. LXXIII)	0.0333	0.0307	8.6%
Σm_ν	Hopf charges on S^3	58.7 meV	< 120 meV	consistent
Hypercharges	topology of K	$+1/6, +2/3, -1/3$	exact	exact
3 generations	$c_1(\mathbb{CP}^2) = 3$	3	3	exact
$S_{\text{BH}} = A/4G_N$	Wald formula	exact	exact	exact

5. The unified equation

5.1 Derivation from the information action

The information action is the length of the path that $\hat{\rho}$ traces in the Fisher metric:

$$\sigma[\mathcal{F}] = \int_K \mathcal{F}_{AB} \dot{\rho}^A \dot{\rho}^B d\tau_{\text{info}} \quad (14)$$

Extremising $\sigma[\mathcal{F}]$ with respect to the trajectory gives the geodesic equation on K with the Fubini-Study metric. On $\mathbb{CP}^2$, the geodesics are precisely the unitary orbits

of $\hat{\rho}$, so the geodesic equation reduces to:

$$i\hbar \frac{d\hat{\rho}}{dt} = [H, \hat{\rho}] \quad (15)$$

The Hamiltonian is uniquely determined to be $H = \mathcal{D}_K^2$ by the consistency condition between the quantum and gravitational sectors: the spectral action of $\mathcal{D}_K^2$ generates both the Yang-Mills and Einstein-Hilbert actions (Connes [6]).

Extremising $\sigma[\mathcal{F}]$ with respect to the 4D metric $g_{\mu\nu}$ gives, via the Bianchi identity, the Einstein equation with the information resistance tensor as source:

The unified equation:

$$\underbrace{i\hbar \frac{d\hat{\rho}}{dt} = [\mathcal{D}_K^2, \hat{\rho}]}_{\text{quantum mechanics}} \longleftrightarrow \underbrace{G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N \mathcal{R}_{\mu\nu}[\mathcal{F}^{(Q)}[\hat{\rho}]]}_{\text{general relativity}} \quad (16)$$

The two equations are not independent. They are two aspects of the single information action $\sigma[\mathcal{F}^{(Q)}]$. The coupling map is: $\hat{\rho} \rightarrow \mathcal{F}^{(Q)}[\hat{\rho}] \rightarrow \mathcal{R}_{\mu\nu}[\mathcal{F}] \rightarrow G_{\mu\nu}$.

5.2 Gravity is already quantised

The Einstein equation (16) is a consequence of the Von Neumann equation, not an independent law. As $\hat{\rho}$ evolves quantum mechanically, the Fisher tensor $\mathcal{F}^{(Q)}[\hat{\rho}]$ — and therefore the spacetime metric $g_{AB} = \mathcal{F}_{AB}/\rho_0$ — changes in step. Spacetime geometry is a quantum observable.

This resolves the problem of quantum gravity not by quantising general relativity but by recognising that it was never classical: both quantum mechanics and gravity are consequences of the single postulate $g_{AB} = \mathcal{F}_{AB}/\rho_0$.

The graviton is the massless zero mode of $\mathcal{D}_K$ at $\lambda = 0$ — part of the information geometry of K , not a separate particle to be quantised.

5.3 The cosmological constant

The fundamental cosmological constant is:

$$\Lambda_{\text{fundamental}} = 0 \quad (17)$$

The observed $\Lambda_{\text{obs}} \approx 10^{-122} M_{\text{Pl}}^2$ is not a fundamental constant but an artefact of measuring distances in ordinary time units rather than informative time units. The informative time element $d\tau_{\text{info}} = \sqrt{\mathcal{F}_{AB} d\rho^A d\rho^B}$ scales with the mean Fisher

information $\bar{\mathcal{F}}^{1/2}$ of the universe. As the universe evolves toward the vacuum state $\hat{\rho}^* = \mathbf{I}/6$, $\bar{\mathcal{F}}$ decreases. Distant supernovae, measured at epochs of higher $\bar{\mathcal{F}}$, appear fainter when measured in current informative time units.

6. Falsifiable predictions

Prediction 1 (Euclid/CMB-S4, 2030):

$$\Sigma m_\nu \approx 58.7 \text{ meV} \quad (18)$$

From the Hopf charge structure of the S^3 lepton sector (Document LXXIII): $\Delta m_{21}^2 / \Delta m_{32}^2 = \tau / (2n_{\text{gen}}) = \tau / 6$, with $m_1 \approx 0$. A measurement of $\Sigma m_\nu < 30 \text{ meV}$ **falsifies** the framework.

Prediction 2 (available now):

$$\sigma_{xy} = \frac{3e^2}{h} \quad (19)$$

Quantised Hall conductance in kagomé metals from the qutrit topological structure of $\mathbb{CP}^2$.

Prediction 3 (LiteBIRD, 2028):

$$n_s = 0.964 \quad (20)$$

Primordial spectral index from information-geometric inflation.

Prediction 4:

$$T_c = \frac{3\hbar}{2k_B\tau_c} \quad (21)$$

Superconducting critical temperature from the information geometry of the vacuum.

7. Open questions

The framework is structurally complete. The remaining open points are calculational:

1. **Explicit KK reduction.** Show term by term that the energy-momentum tensor of 11D SUGRA after reduction on $M^{1,1,1}$ has the structure $8\pi G_N \mathcal{R}_{\mu\nu}[\mathcal{F}^{(Q)}]$.
2. **PMNS matrix.** The lepton mixing matrix from the same Killing-spinor machinery that gives CKM.

3. **CP phase from Chern-Simons.** Derive $\delta = \arctan \varphi^2$ from the Chern-Simons structure of K directly; currently established algebraically but not from first principles.
4. G_N **quantitatively.** Newton's constant as an explicit function of $\mathcal{F}_{AB}$ and M_c .
5. **Wolfenstein A .** The parameter $A = |V_{cb}|/\lambda_W^2 = 0.891$ deviates 8% from the observed 0.826; a subleading correction in the KK expansion is needed.

8. Discussion

The FisherGeometrics framework is not a modification of the Standard Model or of general relativity. It is a rederivation of both from a more fundamental principle: the geometry of distinguishability.

The postulate $g_{AB} = \mathcal{F}_{AB}/\rho_0$ has three immediate consequences. First, quantisation is not an external procedure: the Fisher geometry of the state space is the quantum geometry (Braunstein- Caves). Second, the internal space K is uniquely determined by the quantum information content of the SM vacuum. Third, the Einstein equation follows from the information action by variational principle, with no $T_{\mu\nu}$ input.

The agreement with observation across twelve independent quantities — spanning particle physics, nuclear physics, and cosmology — with zero free parameters, is difficult to attribute to coincidence.

The framework is falsifiable: a single measurement of $\Sigma m_\nu < 30$ meV would refute it. The predicted value $\Sigma m_\nu \approx 58.7$ meV (Document LXXIII) is derived from the Hopf charge structure of the lepton sector on S^3 , with four inputs: $\tau = 1/5$, $n_{\text{gen}} = 3$, the observed Δm_{32}^2 , and the Hopf charges $q = 0, 1, 2$ of the three Killing spinors.

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